

Calculus I

Trigonometry review

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Outline

1 Trigonometry

- Angles
- The Trigonometric Functions
- Trigonometric Identities
- Graphs of the Trigonometric Functions

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Degrees and radians

- Degrees is a unit for measuring angles, denoted by $^{\circ}$.
- The relationship between degrees and radians is:

$$\pi \text{ rad} = 180^{\circ}$$

$$1 \text{ rad} = \frac{180^{\circ}}{\pi} \approx 57.3^{\circ}$$

$$1^{\circ} = \frac{\pi}{180} \text{ rad} \approx 0.017 \text{ rad}.$$

- In other words, a half-turn is measured by $\pi \text{ rad}$ or 180° .
- Degrees are useful because the most frequently encountered fractions of a half turn are measured by a whole number of degrees.
- If a measurement unit is not specified, it is implied to be radians. For example, in $\sin 5$, the number 5 stands for 5 radians.

- Integer quantities of radians are not rational multiples of (the measure of) a half-turn and are not easy to compute with.
- For example to determine in which quadrant is an angle of k radians located one needs to know the numerical value of $\frac{k}{\pi}$, which requires knowledge of π with great numerical accuracy.

Trigonometric Functions and Right Angle Triangles

$\cos \theta = \frac{x}{r}$	$\sec \theta = \frac{r}{x}$	$\cos \theta = \frac{\text{adj}}{\text{hyp}}$	$\sec \theta = \frac{\text{hyp}}{\text{adj}}$
$\sin \theta = \frac{y}{r}$	$\csc \theta = \frac{r}{y}$	$\sin \theta = \frac{\text{opp}}{\text{hyp}}$	$\csc \theta = \frac{\text{hyp}}{\text{opp}}$
$\tan \theta = \frac{y}{x}$	$\cot \theta = \frac{x}{y}$	$\tan \theta = \frac{\text{opp}}{\text{adj}}$	$\cot \theta = \frac{\text{adj}}{\text{opp}}$
All angles		Acute angles	

- The trigonometric functions can be defined without requesting that the pt. (x, y) on the terminal arm of the angle lie on the unit circle.
- To do so we rescale by the distance r from the origin.
- The trig functions of acute θ (between 0 and $\frac{\pi}{2}$) can be interpreted as ratios of sides of right angle triangle with angle θ .

Geometric interpretation of all trigonometric functions

Fix unit circle, center O , coordinates $(0, 0)$.
 Let $\angle DOB = \theta$. Let OB intersect the circle at point A . Coordinates of A are $(\cos \theta, \sin \theta)$.

$$\begin{aligned}\beta &= 180^\circ - 90^\circ - \theta \\ &= 90^\circ - \theta\end{aligned}$$

$$\begin{aligned}\angle OED &= 180^\circ - 90^\circ - \beta \\ &= 90^\circ - (90^\circ - \theta) \\ &= \theta\end{aligned}$$

$$\begin{aligned}\sin \theta &= \frac{\text{opp}}{\text{hyp}} = \frac{|AC|}{|OA|} = \frac{|AC|}{1} = |AC| \\ \cos \theta &= \frac{\text{adj}}{\text{hyp}} = \frac{|OC|}{|OA|} = \frac{|OC|}{1} = |OC| \\ \tan \theta &= \frac{\text{opp}}{\text{adj}} = \frac{|BD|}{|OD|} = \frac{|BD|}{1} = |BD| \\ \cot \theta &= \frac{\text{adj}}{\text{opp}} = \frac{|DE|}{|OD|} = \frac{|DE|}{1} = |DE| \\ \sec \theta &= \frac{\text{hyp}}{\text{adj}} = \frac{|OB|}{|OD|} = \frac{|OB|}{1} = |OB| \\ \csc \theta &= \frac{\text{hyp}}{\text{opp}} = \frac{|OE|}{|DO|} = \frac{|OE|}{1} = |OE|\end{aligned}$$

Example

Find the exact values of the trigonometric functions of

$$\theta = \frac{2\pi}{3} = 120^\circ.$$

$$\begin{aligned} \sin\left(\frac{2\pi}{3}\right) &= \frac{\sqrt{3}}{2} & \cos\left(\frac{2\pi}{3}\right) &= -\frac{1}{2} & \tan\left(\frac{2\pi}{3}\right) &= \frac{\sqrt{3}}{-1} = -\sqrt{3} \\ \csc\left(\frac{2\pi}{3}\right) &= \frac{2}{\sqrt{3}} & \sec\left(\frac{2\pi}{3}\right) &= -\frac{2}{1} = -2 & \cot\left(\frac{2\pi}{3}\right) &= -\frac{1}{\sqrt{3}} \end{aligned}$$

Example

If $\cos \theta = \frac{2}{5}$ and $0 < \theta < \frac{\pi}{2}$, find the other five trigonometric functions of θ .

- Label the hypotenuse with length 5 and the adjacent side with length 2.
- Pythagorean theorem: $x^2 + 2^2 = 5^2$.
- Therefore $x^2 = 21$, so $x = \sqrt{21}$.

$$\sin \theta = \frac{\sqrt{21}}{5} \quad \tan \theta = \frac{\sqrt{21}}{2}$$

$$\csc \theta = \frac{5}{\sqrt{21}} \quad \sec \theta = \frac{5}{2}$$

$$\cot \theta = \frac{2}{\sqrt{21}}$$

$$\begin{array}{ll} \sin \theta = \frac{y}{r} & \csc \theta = \frac{r}{y} \\ \cos \theta = \frac{x}{r} & \sec \theta = \frac{r}{x} \\ \tan \theta = \frac{y}{x} & \cot \theta = \frac{x}{y} \end{array}$$

- Positive angles are obtained by rotating counterclockwise.
- Negative angles are obtained by rotating clockwise.
- If (x, y) is on the terminal arm of the angle θ , then $(x, -y)$ is on the terminal arm of $-\theta$.
- $\sin(-\theta) = \frac{-y}{r} = -\frac{y}{r} = -\sin \theta$.
- $\cos(-\theta) = \frac{x}{r} = \cos \theta$.
- $\sin$ is an odd function.
- $\cos$ is an even function.

$$\begin{array}{ll} \sin \theta = \frac{y}{r} & \csc \theta = \frac{r}{y} \\ \cos \theta = \frac{x}{r} & \sec \theta = \frac{r}{x} \\ \tan \theta = \frac{y}{x} & \cot \theta = \frac{x}{y} \end{array}$$

- 2π represents a full rotation.
- $\theta + 2\pi$ has the same terminal arm as θ .
- $\theta + 2\pi$ uses the same point (x, y) and the same length r .
- $\sin(\theta + 2\pi) = \sin \theta$.
- $\cos(\theta + 2\pi) = \cos \theta$.
- We say $\sin$ and $\cos$ are 2π -periodic.

Trigonometric Identities

Definition (Trigonometric Identity)

A trigonometric identity is an equality between the trigonometric functions in one or more variables that holds for all values of the involved variables in the domains of all of the expressions.

- By convention, when dealing with trigonometric identities we do not account for the domains of the involved expressions.
- For example, $\frac{\sin \theta}{\sin \theta} = 1$ is considered a valid trigonometric identity, although, when considered as a function, the left hand side is not defined for $\theta \neq 0$.

$$\begin{aligned}\sin \theta &= \frac{y}{r} & \csc \theta &= \frac{r}{y} \\ \cos \theta &= \frac{x}{r} & \sec \theta &= \frac{r}{x} \\ \tan \theta &= \frac{y}{x} & \cot \theta &= \frac{x}{y}\end{aligned}$$

- $\csc \theta = \frac{1}{\sin \theta}$
- $\sec \theta = \frac{1}{\cos \theta}$
- $\cot \theta = \frac{1}{\tan \theta}$
- $\tan \theta = \frac{\sin \theta}{\cos \theta}$
- $\cot \theta = \frac{\cos \theta}{\sin \theta}$

$$\begin{aligned}\sin \theta &= \frac{y}{r} & \csc \theta &= \frac{r}{y} \\ \cos \theta &= \frac{x}{r} & \sec \theta &= \frac{r}{x} \\ \tan \theta &= \frac{y}{x} & \cot \theta &= \frac{x}{y}\end{aligned}$$

$$\begin{aligned}& \sin^2 \theta + \cos^2 \theta \\&= \frac{y^2}{r^2} + \frac{x^2}{r^2} \\&= \frac{y^2 + x^2}{r^2} \\&= \frac{r^2}{r^2} \\&= 1\end{aligned}$$

Therefore $\sin^2 \theta + \cos^2 \theta = 1$.

$$\begin{array}{ll} \sin \theta = \frac{y}{r} & \csc \theta = \frac{r}{y} \\ \cos \theta = \frac{x}{r} & \sec \theta = \frac{r}{x} \\ \tan \theta = \frac{y}{x} & \cot \theta = \frac{x}{y} \end{array}$$

Example ($\tan^2 \theta + 1 = \sec^2 \theta$)

Prove the identity

$$\tan^2 \theta + 1 = \sec^2 \theta.$$

$$\begin{aligned} \sin^2 \theta + \cos^2 \theta &= 1 \\ \frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\cos^2 \theta} &= \frac{1}{\cos^2 \theta} \\ \tan^2 \theta + 1 &= \sec^2 \theta \end{aligned}$$

The remaining identities are consequences of the addition formulas:

$$\begin{aligned}\sin(x + y) &= \sin x \cos y + \cos x \sin y \\ \cos(x + y) &= \cos x \cos y - \sin x \sin y\end{aligned}$$

Substitute $-y$ for y , and use the fact that $\sin(-y) = -\sin y$ and $\cos(-y) = \cos y$:

$$\begin{aligned}\sin(x - y) &= \sin x \cos y - \cos x \sin y \\ \cos(x - y) &= \cos x \cos y + \sin x \sin y\end{aligned}$$

The remaining identities are consequences of the addition formulas:

$$\sin(x + y) = \sin x \cos y + \cos x \sin y$$

$$\cos(x + y) = \cos x \cos y - \sin x \sin y$$

To get the double angle formulas, substitute x for y :

$$\sin(2x) = 2 \sin x \cos x$$

$$\cos(2x) = \cos^2 x - \sin^2 x$$

Rewrite the second double angle formula in two ways, using $\cos^2 x = 1 - \sin^2 x$ and $\sin^2 x = 1 - \cos^2 x$:

$$\cos(2x) = 2 \cos^2 x - 1$$

$$\cos(2x) = 1 - 2 \sin^2 x$$

To get the half-angle formulas, solve these equations for $\cos^2 x$ and $\sin^2 x$ respectively.

$$\cos^2 x = \frac{1 + \cos(2x)}{2}, \quad \sin^2 x = \frac{1 - \cos(2x)}{2}$$

The remaining identities are consequences of the addition formulas:

$$\begin{aligned}\sin(x + y) &= \sin x \cos y + \cos x \sin y \\ \cos(x + y) &= \cos x \cos y - \sin x \sin y\end{aligned}$$

Divide the first equation by the second, and then cancel $\cos x \cos y$ from the top and bottom:

$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

Do the same for the subtraction formulas:

$$\tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

Graph of $\sin x$

x	0	$\frac{\pi}{6}$	$\frac{\pi}{2}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{3\pi}{2}$	$\frac{11\pi}{6}$	2π
$\sin x$	0	$\frac{1}{2}$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0

The graph of $\sin x$ is 2π -periodic so the rest of the graph can be inferred from the interval $[0, 2\pi]$.

Graph of $\cos x$

x	0	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	2π
$\cos x$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1

The graph of $\cos x$ is 2π -periodic so the rest of the graph can be inferred from the interval $[0, 2\pi]$.

Graph of $\tan x$

Near $\pm\frac{\pi}{2}$ the graph of $\tan x$ approaches $\pm\infty$. The graph of $\tan x$ is π -periodic so the rest of the graph can be inferred from the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$.

Graphs of the Trigonometric Functions

$$y = \sin x$$

$$y = \cos x$$

- $\sin x$ has zeroes at $n\pi$ for all integers n .
- $\cos x$ has zeroes at $\frac{\pi}{2} + n\pi$ for all integers n .
- $-1 \leq \sin x \leq 1$.
- $-1 \leq \cos x \leq 1$.
- If we translate the graph of $\cos x$ by $\frac{\pi}{2}$ units to the right we get the graph of $\sin x$. This is a consequence of $\cos\left(x - \frac{\pi}{2}\right) = \sin x$.

$$y = \tan x \quad y = \cot x$$

If we move the graph of $\tan x$ by $\frac{\pi}{2}$ units to the left (or right) and reflect across the x axis, we get the graph of $\cot x$. This follows from $\tan\left(x \pm \frac{\pi}{2}\right) = -\cot x$.

$$y = \tan x \quad y = \cot x$$

If we move the graph of $\tan x$ by $\frac{\pi}{2}$ units to the left (or right) and reflect across the x axis, we get the graph of $\cot x$. This follows from $\tan\left(x \pm \frac{\pi}{2}\right) = -\cot x$.

$$y = \csc X \quad y = \sec X$$